1) a)

$\sqrt{-9} = \sqrt{9} \times \sqrt{-1} = 3\text{i}$

1) b)

$\sqrt{-2} = \sqrt{2} \times \sqrt{-1} = \sqrt{2}\text{i}$

1) c)

$2\sqrt{-4} = 2 \times \sqrt{4} \times \sqrt{-1} = 4\text{i}$

1) d)

$-3\sqrt{-2} = -3 \times \sqrt{2} \times \sqrt{-1} = -3\sqrt{2}\text{i}$

2)

$\sqrt{-a} \times \sqrt{-b} = \sqrt{a}\text{i} \times \sqrt{b}\text{i} = \sqrt{a \times b}\text{i}^2 = -\sqrt{ab}$

3) a)

$\text{i}^2 = -1$

3) b)

$\text{i}^3 = \text{i} \times \text{i}^2 = \text{i} \times -1 = -\text{i}$

3) c)

$\text{i}^4 = \text{i}^2 \times \text{i}^2 = -1 \times -1 = 1$

3) d)

$\text{i}^5 = \text{i} \times (\text{i}^2)^2 = \text{i} \times (-1)^2 = \text{i}$

3) e)

$\text{i}^{40} = (\text{i}^2)^{20} = (-1)^{20} = 1$

3) f)

$\text{i}^{41} = \text{i} \times (\text{i}^2)^{20} = \text{i} \times (-1)^{20} = \text{i}$

3) g)

$i^{99} = i \times (i^2)^{49} = i \times (-1)^{49} = -i$

4) a)

$i^{4n} = (i^2)^{2n} = (-1)^{2n} = 1$

4) b)

$i^{4n+1} = i \times i^{4n} = i \times (i^2)^{2n} = i \times (-1)^{2n} = i$

4) c)

$i^{4n+2} = i^2 \times i^{4n} = i^2 \times (i^2)^{2n} = -1 \times (-1)^{2n} = -1$

4) d)

$i^{4n+3} = i^3 \times i^{4n} = i \times i^2 \times (i^2)^{2n} = i \times -1 \times (-1)^{2n} = -i$

4) e)

$i^{4n+17} = i \times i^{4n+16} = i \times (i^2)^{2n+8} = i \times (-1)^{2n+8} = i$

4) f)

$i^{4n+26} = i^2 \times i^{4n+24} = -1 \times (i^4)^{n+6} = -1 \times (1)^{n+6} = -1$

5)

$i + i^2 + i^3 + i^4 = i - 1 - i + 1 = 0$

$i + i^2 + i^3 + \cdots + i^{1000} = 250(i - 1 - i + 1) = 0$

6) a)

$(1 - i)^2 = (1 - i)(1 - i) = 1 - 2i + (-i)^2 = 1 - 2i - 1 = -2i$

6) b)

$(1 - i)^3 = (1 - i)(1 - i)(1 - i) = -2i(1 - i) = -2i + 2i^2 = -2 - 2i$

6) c)

$(1 - i)^2 = (1 - i)(1 - i) = 1 - 2i + (-i)^2 = 1 - 2i - 1 = -2i$

$(1 - i)^4 = (1 - i)^2(1 - i)^2 = -2i \times -2i = 4 \times -1 = -4$

6) d)

$(1+\mathrm{i})^2 = (1+\mathrm{i})(1+\mathrm{i}) = 1+2\mathrm{i}+\mathrm{i}^2 = 2\mathrm{i}$

6) e)

$(1+\mathrm{i})^3 = (1+\mathrm{i})(1+\mathrm{i})(1+\mathrm{i}) = 2\mathrm{i}(1+\mathrm{i}) = 2\mathrm{i}+2\mathrm{i}^2 = -2+2\mathrm{i}$

6) f)

$(1+\mathrm{i})^2 = (1+\mathrm{i})(1+\mathrm{i}) = 1+2\mathrm{i}+\mathrm{i}^2 = 2\mathrm{i}$

$(1+\mathrm{i})^4 = (1+\mathrm{i})^2(1+\mathrm{i})^2 = 2\mathrm{i} \times 2\mathrm{i} = 4 \times -1 = -4$

7) a)

$\sqrt{-12} \times \sqrt{-27} = \sqrt{3 \times 4 \times -1} \times \sqrt{3 \times 9 \times -1} = 2\sqrt{3}\mathrm{i} \times 3\sqrt{3}\mathrm{i} = 6 \times 3 \times -1 = -18$

7) b)

$(2\mathrm{i})^3 = 8\mathrm{i}^3 = -8\mathrm{i}$

7) c)

$\mathrm{i}^2(\mathrm{i}+3) = -1(\mathrm{i}+3) = -\mathrm{i}-3$

7) d)

$\mathrm{i}+4\mathrm{i}^5 = \mathrm{i}+4(-1)(-1)\mathrm{i} = \mathrm{i}+4\mathrm{i} = 5\mathrm{i}$

7) e)

$(\mathrm{i}-3\mathrm{i}^3)(1-2\mathrm{i}^4) = (\mathrm{i}+3\mathrm{i})(1-2) = (4\mathrm{i})(-1) = -4\mathrm{i}$

7) f)

$(2\mathrm{i}^6-\mathrm{i}^9)^2 = (2(-1)^3-\mathrm{i}(-1)^4)^2 = (-2-\mathrm{i})^2 = 4+4\mathrm{i}+\mathrm{i}^2 = 3+4\mathrm{i}$

8)

$\left(\mathrm{i}^{12}+\left(\frac{1}{\mathrm{i}}\right)^{18}\right)^3 = \left((-1)^6+\left(\frac{1}{-1}\right)^9\right)^3 = (1-1)^3 = 0$

9)

$\left(\mathrm{i}^{30}-\left(\frac{1}{\mathrm{i}}\right)^{12}\right)^{4}=\left((-1)^{15}-\left(\frac{1}{-1}\right)^{6}\right)^{4}=(-1-1)^{4}=16$

10) a)

$2-3\mathrm{i}+(1+\mathrm{i})=3-2\mathrm{i}$

10) b)

$2-3\mathrm{i}+(5-2\mathrm{i})=7-5\mathrm{i}$

10) c)

$5-2\mathrm{i}-(2-3\mathrm{i})=3+\mathrm{i}$

10) d)

$2(1+\mathrm{i})-(2-3\mathrm{i})=2+2\mathrm{i}-2+3\mathrm{i}=5\mathrm{i}$

10) e)

$5(2-3\mathrm{i})+3(1+\mathrm{i})=10-15\mathrm{i}+3+3\mathrm{i}=13-12\mathrm{i}$

11)

$(2+3\mathrm{i})^{2}=(2+3\mathrm{i})(2+3\mathrm{i})=4+12\mathrm{i}-9=-5+12\mathrm{i}$ as req.

12)

$(-1+5\mathrm{i})^{3}=(-1+5\mathrm{i})(-1+5\mathrm{i})(-1+5\mathrm{i})=(1-10\mathrm{i}-25)(-1+5\mathrm{i})=$
$(-24-10\mathrm{i})(-1+5\mathrm{i})=24+10\mathrm{i}-120\mathrm{i}+50=74-110\mathrm{i}$ as req.

13) a)

$\mathrm{Re}(z)=2+3x$ and $\mathrm{Im}(z)=-y$

13) b)

$\mathrm{Re}(z)=x-2x=-x$ and $\mathrm{Im}(z)=-2-3y$

13) c)

$\mathrm{Re}(z)=5x+2x=7x$ and $\mathrm{Im}(z)=-5y+2y=-3y$

14) a)

$5x + 2y = 25$ and $3x + 4y = 29$

Solving simultaneously: $7x = 21 \therefore x = 3$ and $y = 5$

14) b)

$2x + 8y = -20$ and $x - 4y = 14$

Solving simultaneously: $4x = 8 \therefore x = 2$ and $y = -3$

14) c)

$(-3x + 2\mathrm{i}x) + (2y + 7\mathrm{i}y) = -3x + 2y + (2x + 7y)\mathrm{i}$

$-3x + 2y + (2x + 7y)\mathrm{i} = -14 - 74\mathrm{i}$

$-3x + 2y = -14$ and $2x + 7y = -74$

Solving simultaneously: $25y = -250 \therefore x = -2$ and $y = -10$

14) d)

$x + 3x^2 = 2$ and $9x - y = 1$

$3x^2 + x - 2 = 0 \Rightarrow x = \frac{-1 \pm \sqrt{1+24}}{6} = \frac{2}{3}$ or -1

$\therefore x = \frac{2}{3}$ and $y = 5$ or $x = -1$ and $y = -10$

15)

$z = (3 + 7\mathrm{i})(a + 2\mathrm{i}) = 3a + 6\mathrm{i} + 7a\mathrm{i} - 14$

$\mathrm{Re}(z) = 3a - 14 = 0 \therefore a = \frac{14}{3}$

16)

$z = (2 + \mathrm{i})(a + b\mathrm{i}) = 2a + a\mathrm{i} + 2b\mathrm{i} - b$

$\mathrm{Re}(z) = 0 \therefore 2a - b = 0$ so $b = 2a$

$z = 2a + a\mathrm{i} + 2(2a)\mathrm{i} - 2a = 5a\mathrm{i}$

17)

$(a - b\mathrm{i})(a + 3b\mathrm{i}) = a^2 + 3ab\mathrm{i} - ab\mathrm{i} + 3b^2 = a^2 + 3b^2 + 2ab\mathrm{i}$

$a^2 + 3b^2 = 52$ and $2ab = 16$

$\therefore a = \frac{8}{b}$ so $\left(\frac{8}{b}\right)^2 + 3b^2 = 52$

$\frac{64}{b^2} + 3b^2 = 52 \Rightarrow 64 + 3b^4 = 52b^2 \Rightarrow 3b^4 - 52b^2 + 64 = 0$

$\therefore b^2 = \frac{4}{3}$ or $b^2 = 16$

Since b is an integer, $b = \pm 4$

$\therefore a = -2$ and $b = -4$ or $a = 2$ and $b = 4$

18) a)

$(z + 1 - 2\mathrm{i})^2 = (z + 1 - 2\mathrm{i})(z + 1 - 2\mathrm{i})$

$= z^2 + z - 2\mathrm{i}z + z + 1 - 2\mathrm{i} - 2\mathrm{i}z - 2\mathrm{i} - 4$

$= z^2 + 2z - 4\mathrm{i}z - 3 - 4\mathrm{i}$

$= z^2 + (2 - 4\mathrm{i})z + (-3 - 4\mathrm{i})$

18) b)

$(z - 2 + 3\mathrm{i})^2 = (z - 2 + 3\mathrm{i})(z - 2 + 3\mathrm{i})$

$= z^2 - 2z + 3\mathrm{i}z - 2z + 4 - 6\mathrm{i} + 3\mathrm{i}z - 6\mathrm{i} - 9$

$= z^2 - 4z + 6\mathrm{i}z - 5 - 12\mathrm{i}$

$= z^2 + (-4 + 6\mathrm{i})z + (-5 - 12\mathrm{i})$

18) c)

$(z + 1 + 2\mathrm{i})(z - 2 + 3\mathrm{i})$

$= z^2 - 2z + 3\mathrm{i}z + z + 2 + 3\mathrm{i} + 2\mathrm{i}z - 4\mathrm{i} + 6$

$= z^2 - z + 5\mathrm{i}z - 8 - \mathrm{i}$

$= z^2 + (-1 + 5\mathrm{i})z + (-8 - \mathrm{i})$

19) a)

$(7 + 2\mathrm{i})(3 + 8\mathrm{i}) = 21 + 56\mathrm{i} + 6\mathrm{i} - 16 = 5 + 62\mathrm{i}$

19) b)

$(7 - 2\mathrm{i})(3 - 8\mathrm{i}) = 21 - 6\mathrm{i} - 56\mathrm{i} - 16 = 5 - 62\mathrm{i}$

19) c)

$5^2 + 62^2 = (5 + 62\text{i})(5 - 62\text{i})$

$= (7 + 2\text{i})(7 - 2\text{i})(3 + 8\text{i})(3 - 8\text{i})$

Considering the real part (the imaginary part is zero):

$\text{Re}\big((7 + 2\text{i})(7 - 2\text{i})(3 + 8\text{i})(3 - 8\text{i})\big)$

$= (49 + 4)(9 + 64) = 53 \times 73$

20) a)

Let $z = a + b\text{i}$ so $z^2 = a^2 - b^2 + 2ab\text{i}$

$a^2 - b^2 + 2ab\text{i} = 5 + 12\text{i}$

Compare real parts: $a^2 - b^2 = 5$

Compare imaginary parts: $2ab = 12$

Solve simultaneously:

$b = \frac{6}{a}$ gives $a^2 - \left(\frac{6}{a}\right)^2 = 5 \Rightarrow a^4 - 36 = 5a^2 \Rightarrow a^4 - 5a^2 - 36 = 0$

$a^2 = \frac{5 \pm \sqrt{25+144}}{2} = \frac{5}{2} \pm \frac{13}{2}$ so $a^2 = 9$ or $a^2 = -4$

Since a is real, $a = -3$ or $a = 3$

$\therefore a = -3, b = -2$ or $a = 3, b = 2$

$z = -3 - 2\text{i}$ or $z = 3 + 2\text{i}$

20) b)

Let $z = a + b\text{i}$ so $z^2 = a^2 - b^2 + 2ab\text{i}$

$a^2 - b^2 + 2ab\text{i} = 21 - 20\text{i}$

Compare real parts: $a^2 - b^2 = 21$

Compare imaginary parts: $2ab = -20$

Solve simultaneously:

$b = -\frac{10}{a}$ gives $a^2 - \left(-\frac{10}{a}\right)^2 = 21 \Rightarrow a^4 - 100 = 21a^2 \Rightarrow a^4 - 21a^2 - 100 = 0$

$a^2 = \frac{21 \pm \sqrt{441+400}}{2} = \frac{21}{2} \pm \frac{29}{2}$ so $a^2 = 25$ or $a^2 = -4$

Since a is real, $a = -5$ or $a = 5$

$\therefore a = -5, b = 2$ or $a = 5, b = -2$

$z = -5 + 2\text{i}$ or $z = 5 - 2\text{i}$

20) c)

Let $z = a + b\text{i}$ so $z^2 = a^2 - b^2 + 2ab\text{i}$

$a^2 - b^2 + 2ab\text{i} = \frac{3}{16} - \frac{1}{4}\text{i}$

Compare real parts: $a^2 - b^2 = \frac{3}{16}$

Compare imaginary parts: $2ab = -\frac{1}{4}$

Solve simultaneously:

$b = -\frac{1}{8a}$ gives $a^2 - \left(-\frac{1}{8a}\right)^2 = \frac{3}{16} \Rightarrow a^4 - \frac{1}{64} = \frac{3}{16}a^2 \Rightarrow 64a^4 - 12a^2 - 1 = 0$

$a^2 = \frac{12 \pm \sqrt{144+256}}{128} = \frac{12}{128} \pm \frac{20}{128}$ so $a^2 = \frac{1}{4}$ or $a^2 = -\frac{1}{16}$

Since a is real, $a = -\frac{1}{2}$ or $a = \frac{1}{2}$

$\therefore a = -\frac{1}{2}, b = \frac{1}{4}$ or $a = \frac{1}{2}, b = -\frac{1}{4}$

$z = -\frac{1}{2} + \frac{1}{4}\text{i}$ or $z = \frac{1}{2} - \frac{1}{4}\text{i}$

21) a)

$z = \sqrt{4 + 3\text{i}} \Rightarrow z^2 = 4 + 3\text{i}$

Let $z = a + b\text{i}$ so $z^2 = a^2 - b^2 + 2ab\text{i}$

$a^2 - b^2 + 2ab\text{i} = 4 + 3\text{i}$

Compare real parts: $a^2 - b^2 = 4$

Compare imaginary parts: $2ab = 3$

Solve simultaneously:

$b = \frac{3}{2a}$ gives $a^2 - \left(\frac{3}{2a}\right)^2 = 4 \Rightarrow a^4 - \frac{9}{4} = 4a^2 \Rightarrow 4a^4 - 16a^2 - 9 = 0$

$a^2 = \frac{16 \pm \sqrt{256+144}}{8} = 2 \pm \frac{20}{8}$ so $a^2 = \frac{9}{2}$ or $a^2 = -\frac{1}{2}$

Since a is real, $a = \frac{3\sqrt{2}}{2}$ or $a = -\frac{3\sqrt{2}}{2}$

$\therefore a = \frac{3\sqrt{2}}{2}, b = \frac{\sqrt{2}}{2}$ or $a = -\frac{3\sqrt{2}}{2}, b = -\frac{\sqrt{2}}{2}$

$\therefore \sqrt{4 + 3\text{i}} = \frac{3\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\text{i}$

21) b)

Let $z = a + b\text{i}$ so $z^2 = a^2 - b^2 + 2ab\text{i}$

$a^2 - b^2 + 2ab\text{i} = -12 + 16\text{i}$

Compare real parts: $a^2 - b^2 = -12$

Compare imaginary parts: $2ab = 16$

Solve simultaneously:

$b = \frac{8}{a}$ gives $a^2 - \left(\frac{8}{a}\right)^2 = -12 \Rightarrow a^4 - 64 = -12a^2 \Rightarrow a^4 + 12a^2 - 64 = 0$

$a^2 = \frac{-12 \pm \sqrt{144+256}}{2} = -6 \pm 10$ so $a^2 = 4$ or $a^2 = -16$

Since a is real, $a = 2$ or $a = -2$

$\therefore a = 2, b = 4$ or $a = -2, b = -4$

$\therefore \sqrt{-12 + 16\text{i}} = 2 + 4\text{i}$

21) c)

$z = \sqrt{2 + 4\sqrt{2}\text{i}} \Rightarrow z^2 = 2 + 4\sqrt{2}\text{i}$

Let $z = a + b\text{i}$ so $z^2 = a^2 - b^2 + 2ab\text{i}$

$a^2 - b^2 + 2ab\text{i} = 2 + 4\sqrt{2}\text{i}$

Compare real parts: $a^2 - b^2 = 2$

Compare imaginary parts: $2ab = 4\sqrt{2}$

Solve simultaneously:

$b = \frac{2\sqrt{2}}{a}$ gives $a^2 - \left(\frac{2\sqrt{2}}{a}\right)^2 = 2 \Rightarrow a^4 - 8 = 2a^2 \Rightarrow a^4 - 2a^2 - 8 = 0$

$a^2 = \frac{2 \pm \sqrt{4+32}}{2} = 1 \pm 3$ so $a^2 = 4$ or $a^2 = -2$

Since a is real, $a = 2$ or $a = -2$

$\therefore a = 2, b = \sqrt{2}$ or $a = -2, b = -\sqrt{2}$

$\therefore \sqrt{2 - \text{i}} = 2 + \sqrt{2}\text{i}$

22)

Let $z = a + b\text{i}$ and $w = c + d\text{i}$

$2\text{i}(a + b\text{i}) + 4(c + d\text{i}) = -6 + 8\text{i}$

$\Rightarrow 2a + 4d = 8$ and $-2b + 4c = -6$

$-3(a + b\text{i}) + 5\text{i}(c + d\text{i}) = 7 - \text{i}$

$\Rightarrow -3a - 5d = 7$ and $-3b + 5c = -1$

So

$2a + 4d = 8$ and $-3a - 5d = 7$ gives $a = -34$ and $d = 19$

$-2b + 4c = -6$ and $-3b + 5c = -1$ gives $b = -13$ and $c = -8$

So $z = -34 - 13\text{i}$ and $w = -8 + 19\text{i}$

Alternative Method:

Let $z = a + b\text{i}$ and $w = c + d\text{i}$

$2\text{i}(a + b\text{i}) + 4(c + d\text{i}) = -6 + 8\text{i}$ (1)

$-3(a + b\text{i}) + 5\text{i}(c + d\text{i}) = 7 - \text{i}$ (2)

Multiply (1) by 3

$6\text{i}(a + b\text{i}) + 12(c + d\text{i}) = -18 + 24\text{i}$ (3)

Multiply (2) by 2i

$-6\text{i}(a + b\text{i}) - 10(c + d\text{i}) = 2 + 14\text{i}$ (4)

(3) + (4)

$2(c + d\text{i}) = -16 + 38\text{i}$

$\therefore c = -8$ and $d = 19$

Multiply (1) by 5

$10\text{i}(a + b\text{i}) + 20(c + d\text{i}) = -30 + 40\text{i}$ (5)

Multiply (2) by 4i

$-12\text{i}(a + b\text{i}) - 20(c + d\text{i}) = 4 + 28\text{i}$ (6)

(5) + (6)

$-2\text{i}(a + b\text{i}) = -26 + 68\text{i}$

$-2a\text{i} + 2b = -26 + 68\text{i}$

$\therefore a = -34$ and $b = -13$

So $z = -34 - 13\text{i}$ and $w = -8 + 19\text{i}$

23) a)

$(1+ai)^3 = 1+3ai+3(ai)^2+(ai)^3 = 1+3ai-3a^2-a^3i$

$\therefore \text{Re}((1+ai)^3) = 1-3a^2$

$1-3a^2=0 \Rightarrow 3a^2=1 \Rightarrow a^2=\frac{1}{3} \Rightarrow a=\pm\sqrt{\frac{1}{3}}=\pm\frac{\sqrt{3}}{3}$

23) b)

$(3+ai)^3 = 27+27ai+9(ai)^2+(ai)^3 = 27+27ai-9a^2-a^3i$

$\therefore \text{Im}((3+ai)^3) = 27a-a^3$

$27a-a^3=0 \Rightarrow a(27-a^2)=0 \Rightarrow a=0 \text{ or } a=\pm 3\sqrt{3}$

23) c)

$(4+ai)^4 = 16+32ai+24(ai)^2+8(ai)^3+(ai)^4 = 256+256ai-96a^2-16a^3i+a^4$

$\therefore \text{Im}((4+ai)^4) = 256a-16a^3$

$256a-16a^3=0 \Rightarrow 16a(16-a^2)=0 \therefore a=0, a=-4 \text{ or } a=4$

23) d)

$(2-ai)^4 = 16-32ai+24(-ai)^2+8(-ai)^3+(-ai)^4$

$= 16-32ai-24a^2+8a^3i+a^4$

$\therefore \text{Re}((2-ai)^4) = 16-24a^2+a^4$

$a^4-24a^2+16=0 \Rightarrow a^2=\frac{24\pm\sqrt{576-64}}{2}=12\pm 8\sqrt{2}$

$\therefore a=\sqrt{12+8\sqrt{2}}, a=-\sqrt{12+8\sqrt{2}}, a=\sqrt{12-8\sqrt{2}} \text{ or } a=-\sqrt{12-8\sqrt{2}}$

23) e)

$(1-ai)^5 = 1+5ai+10(-ai)^2+10(-ai)^3+5(-ai)^4+(-ai)^5$

$= 1+5ai-10a^2+10a^3i+5a^4-a^5i$

$\therefore \text{Im}((1-ai)^5) = 5a+10a^3-a^5$

$5a+10a^3-a^5=0 \Rightarrow -a(a^4-10a^2-5)=0$

$\Rightarrow \text{so } a=0 \text{ or } a^2=\frac{10\pm\sqrt{100+20}}{2}=5\pm\sqrt{30}$

$\therefore a=0, a=\sqrt{5+\sqrt{30}} \text{ or } a=\sqrt{5-\sqrt{30}}$

24) a)

$z^* = 2-i$

24) b)

$z^* = 3 + \mathrm{i}$

24) c)

$z^* = -6 + \mathrm{i}$

24) d)

$z^* = 1 - 5\mathrm{i}$

24) e)

$z^* = -4 + 2\mathrm{i}$

24) f)

$z^* = \frac{1}{2} - \frac{3}{4}\mathrm{i}$

24) g)

$z^* = \sqrt{5} + \sqrt{3}\mathrm{i}$

24) h)

$z^* = -2\mathrm{i}$

24) i)

$z^* = -10$

25) a)

$z + z^* = a + b\mathrm{i} + (a - b\mathrm{i}) = 2a$ which is real.

25) b)

$zz^* = (a + b\mathrm{i})(a - b\mathrm{i}) = a^2 + ab\mathrm{i} - ab\mathrm{i} + b^2 = a^2 + b^2$ which is real.

26) a)

$2w - z = 2(2 - \mathrm{i}) - (-5 + 3\mathrm{i}) = 4 - 2\mathrm{i} + 5 - 3\mathrm{i} = 9 - 5\mathrm{i}$

26) b)

$w^* + w = (2 + \mathrm{i}) + (2 - \mathrm{i}) = 4$

26) c)

$2w^* + z = 2(2 + \mathrm{i}) + (-5 + 3\mathrm{i}) = 4 + 2\mathrm{i} - 5 + 3\mathrm{i} = -1 + 5\mathrm{i}$

26) d)

$z^2 = (-5 + 3\mathrm{i})^2 = (-5 + 3\mathrm{i})(-5 + 3\mathrm{i}) = 25 - 15\mathrm{i} - 15\mathrm{i} - 9 = 16 - 30\mathrm{i}$

26) e)

$wzz^* = (2 - \mathrm{i})(-5 + 3\mathrm{i})(-5 - 3\mathrm{i}) = (2 - \mathrm{i})(25 - 15\mathrm{i} + 15\mathrm{i} + 9)$
$= (2 - \mathrm{i}) \times 34$
$= 68 - 34\mathrm{i}$

26) f)

$w(z^*)^2 = (2 - \mathrm{i})(-5 - 3\mathrm{i})^2 = (2 - \mathrm{i})(25 + 15\mathrm{i} + 15\mathrm{i} - 9)$
$= (2 - \mathrm{i})(16 + 30\mathrm{i}) = 32 + 60\mathrm{i} - 16\mathrm{i} + 30$
$= 62 + 44\mathrm{i}$

26) g)

$(w^*)^3 = (2 + \mathrm{i})^3 = (2 + \mathrm{i})(2 + \mathrm{i})(2 + \mathrm{i}) = (2 + \mathrm{i})(4 + 4\mathrm{i} - 1)$
$= (2 + \mathrm{i})(3 + 4\mathrm{i}) = 6 + 8\mathrm{i} + 3\mathrm{i} - 4 = 2 + 11\mathrm{i}$

27) a)

$4\mathrm{i}z + 3z^* = 4\mathrm{i}(x + \mathrm{i}y) + 3(x - \mathrm{i}y) = 4x\mathrm{i} - 4y + 3x - 3\mathrm{i}y = (3x - 4y) + (4x - 3y)\mathrm{i}$
$\mathrm{Re}(4\mathrm{i}z + 3z^*) = 3x - 4y$
$\mathrm{Im}(4\mathrm{i}z + 3z^*) = 4x - 3y$

27) b)

$3x - 4y = 5$ and $4x - 3y = -3$

Solving simultaneously gives $x = -\frac{27}{7}$ and $y = -\frac{29}{7}$

28) a)

$2z^* = 3 - 4\mathrm{i} \Rightarrow z^* = \frac{3}{2} - 2\mathrm{i} \therefore z = \frac{3}{2} + 2\mathrm{i}$

28) b)

$3z^* = -1 + 8\text{i} \Rightarrow z^* = -\frac{1}{3} + \frac{8}{3}\text{i} \therefore z = -\frac{1}{3} - \frac{8}{3}\text{i}$

28) c)

$\text{i}z^* = -1 + \text{i} \Rightarrow -\text{i} \times \text{i}z^* = -\text{i}(-1 + \text{i}) = 1 + \text{i} \therefore z = 1 - \text{i}$

Alternative Method:

$z^* = \frac{-1+\text{i}}{\text{i}} \times \frac{-\text{i}}{-\text{i}} = -\text{i}(-1 + \text{i}) = 1 + \text{i} \therefore z = 1 - \text{i}$

29) a)

Let $z = a + b\text{i}$

$z + 2z^* = 3 + 9\text{i} \Rightarrow a + b\text{i} + 2(a - b\text{i}) = 3 + 9\text{i}$

Compare real parts: $a + 2a = 3 \therefore a = 1$

Compare imaginary parts: $b - 2b = 9 \therefore b = -9$

$\therefore z = 1 - 9\text{i}$

29) b)

Let $z = a + b\text{i}$

$2z - z^* = -4 - 5\text{i} \Rightarrow 2(a + b\text{i}) - (a - b\text{i}) = -4 - 5\text{i}$

Compare real parts: $2a - a = -4 \therefore a = -4$

Compare imaginary parts: $2b + b = -5 \therefore b = -\frac{5}{3}$

$\therefore z = -4 - \frac{5}{3}\text{i}$

29) c)

Let $z = a + b\text{i}$

$z + 5z^* = -\text{i} \Rightarrow a + b\text{i} + 5(a - b\text{i}) = -\text{i}$

Compare real parts: $a + 5a = 0 \therefore a = 0$

Compare imaginary parts: $b - 5b = -1 \therefore b = \frac{1}{4}$

$\therefore z = \frac{1}{4}\text{i}$

30) a)

Let $z = a + b\text{i}$ so $z^* = a - b\text{i}$

$(z^*)^2 = (a - b\text{i})^2 = a^2 - b^2 - 2ab\text{i}$

$(z^2)^* = \big((a + b\text{i})(a + b\text{i})\big)^* = (a^2 - b^2 + 2ab\text{i})^* = a^2 - b^2 - 2ab\text{i}$

$\therefore\ (z^*)^2 = (z^2)^*$ as req.

30) b)

Let $z = a + b\text{i}$ so $z^* = a - b\text{i}$

$(z^*)^3 = (a - b\text{i})^2(a - b\text{i}) = (a^2 - b^2 - 2ab\text{i})(a - b\text{i})$

$= a^3 - ab^2 - 2a^2b\text{i} - a^2b\text{i} + b^3\text{i} - 2ab^2 = a^3 - 3ab^2 + (b^3 - 3a^2b)\text{i}$

$(z^3)^* = \big((a + b\text{i})^2(a + b\text{i})\big)^* = \big((a^2 - b^2 + 2ab\text{i})(a + b\text{i})\big)^*$

$= (a^3 - ab^2 + 2a^2b\text{i} + a^2b\text{i} - b^3\text{i} - 2ab^2)^*$

$= (a^3 - ab^2 - 2ab^2 + (-b^3 + a^2b)\text{i})^* = a^3 - ab^2 - 2ab^2 + (b^3 - a^2b)\text{i}$

$\therefore\ (z^*)^3 = (z^3)^*$ as req.

31) a)

$z - z^* = a + b\text{i} - (a - b\text{i}) = 2b\text{i}$

31) b)

$(z - z^*)^2 = \big(a + b\text{i} - (a - b\text{i})\big)^2 = (2b\text{i})^2 = -4b^2$

31) c)

$(z - z^*)^3 = \big(a + b\text{i} - (a - b\text{i})\big)^3 = (2b\text{i})^3 = -8b^3\text{i}$

31) d)

$(z - z^*)^4 = \big(a + b\text{i} - (a - b\text{i})\big)^4 = (2b\text{i})^4 = 16b^4$

31) e)

$(z - z^*)^{18} = \big(a + b\text{i} - (a - b\text{i})\big)^{18} = (2b\text{i})^{18} = -262144b^{18}$

32)

Let $z = a + b\text{i}$ so $z^* = a - b\text{i}$

$(4z - z^*)^* = \big(4(a + b\text{i}) - (a - b\text{i})\big)^* = (4a + 4b\text{i} - a + b\text{i})^* = (3a + 5b\text{i})^* = 3a - 5b\text{i}$

$z^2 = (a + b\text{i})^2 = a^2 - b^2 + 2ab\text{i}$

Compare real parts: $a^2 - b^2 = 3a$

Compare imaginary parts: $2ab = -5b$

Solve simultaneously:

Consider $2ab = -5b$

If $b \neq 0$, $a = -\frac{5}{2}$ which gives $\left(-\frac{5}{2}\right)^2 - b^2 = 3\left(-\frac{5}{2}\right)$

$\Rightarrow b^2 = \frac{25}{4} + \frac{15}{2} = \frac{55}{4}$ so $b = \frac{\sqrt{55}}{2}$ or $b = -\frac{\sqrt{55}}{2}$

$\therefore a = -\frac{5}{2}, b = \frac{\sqrt{55}}{2}$ or $a = -\frac{5}{2}, b = -\frac{\sqrt{55}}{2}$

If $b = 0$, $a^2 = 3a \Rightarrow a^2 - 3a = 0 \Rightarrow a(a - 3) = 0$

So $a = 0$ or $a = 3$

$\therefore a = 0, b = 0$ or $a = 3, b = 0$

Which gives

$z = 0, z = 3, z = -\frac{5}{2} + \frac{\sqrt{55}}{2}\text{i}$ or $z = -\frac{5}{2} - \frac{\sqrt{55}}{2}\text{i}$

33)

Let $z = a + b\text{i}$, $z^* = a - b\text{i}$ and $w = c + d\text{i}$

$2(a + b\text{i}) + 4(a - b\text{i}) = 18 + 4\text{i}$

$\Rightarrow 2a + 4a = 18$ so $a = 3$ and $2b - 4b = 4$ so $b = -2$

$z = 3 - 2\text{i}$

So $z^* = 3 + 2\text{i}$

$5(c + d\text{i}) - 3(3 + 2\text{i}) = -3 + 2\text{i}$

$\Rightarrow 5c - 9 = -3$, so $c = \frac{6}{5}$ and $5d - 6 = 2$ so $d = \frac{8}{5}$

$\therefore w = \frac{6}{5} + \frac{8}{5}\text{i}$

34)

Let $z = a + b\mathrm{i}$

$(a + b\mathrm{i})^2 + 3\mathrm{i}(a - b\mathrm{i}) = 8$

$a^2 - b^2 + 2ab\mathrm{i} + 3a\mathrm{i} + 3b = 8$

Compare real part: $a^2 - b^2 + 3b = 8$

Compare imaginary part: $2ab + 3a = 0 \Rightarrow a(2b + 3) = 0$

Since $\mathrm{Re}(z) = a < 0$, $b = -\frac{3}{2}$

$a^2 - \left(-\frac{3}{2}\right)^2 + 3\left(-\frac{3}{2}\right) = 8 \Rightarrow a^2 = \frac{59}{4}$

$\therefore a = \frac{\sqrt{59}}{2}$ or $a = -\frac{\sqrt{59}}{2}$

$\therefore z = -\frac{\sqrt{59}}{2} - \frac{3}{2}\mathrm{i}$

35)

Let $z = a + b\mathrm{i}$

$(a + b\mathrm{i})^2 + 4\mathrm{i}(a - b\mathrm{i}) = 10\mathrm{i}$

$a^2 - b^2 + 2ab\mathrm{i} + 4a - 4b\mathrm{i} = 10\mathrm{i}$

Compare real part: $a^2 - b^2 + 4a = -4$

Compare imaginary part: $2ab - 4b = 10 \Rightarrow 2b(a - 2) = 10 \Rightarrow b = \frac{5}{a-2}$

So that $a^2 - b^2 + 4a = -4$ becomes:

$a^2 - \left(\frac{5}{a-2}\right)^2 + 4a = -4$

$a^2(a - 2)^2 - 25 + 4a(a - 2)^2 + 4(a - 2)^2 = 0$

$a^2(a^2 - 4a + 4) - 25 + 4a(a^2 - 4a + 4) + 4(a^2 - 4a + 4) = 0$

$a^4 - 4a^3 + 4a^2 - 25 + 4a^3 - 16a^2 + 16a + 4a^2 - 16a + 16 = 0$

$a^4 - 8a^2 - 9 = 0$

$(a^2 + 1)(a^2 - 9) = 0$

Since a is real $a = \pm 3$

But also $\mathrm{Re}(z) = a > 0$ so $a = 3$

$b = \frac{5}{3-2} = 5$

$\therefore z = 3 + 5\mathrm{i}$

36) a)

$\frac{4}{\mathrm{i}} \times \frac{-\mathrm{i}}{-\mathrm{i}} = -4\mathrm{i}$

36) b)

$$\frac{7}{1+\mathrm{i}} \times \frac{1-\mathrm{i}}{1-\mathrm{i}} = \frac{7-7\mathrm{i}}{1+1} = \frac{7}{2} - \frac{7}{2}\mathrm{i}$$

36) c)

$$\frac{9}{5-2\mathrm{i}} \times \frac{5+2\mathrm{i}}{5+2\mathrm{i}} = \frac{45+18\mathrm{i}}{25+4} = \frac{45}{29} + \frac{18}{29}\mathrm{i}$$

36) d)

$$\frac{8}{2+3\mathrm{i}} \times \frac{2-3\mathrm{i}}{2-3\mathrm{i}} = \frac{16-24\mathrm{i}}{4+9} = \frac{16}{13} - \frac{24}{13}\mathrm{i}$$

36) e)

$$\frac{2+\mathrm{i}}{4-\mathrm{i}} \times \frac{4+\mathrm{i}}{4+\mathrm{i}} = \frac{8+6\mathrm{i}-1}{16+1} = \frac{7}{17} + \frac{6}{17}\mathrm{i}$$

36) f)

$$\frac{(1-\mathrm{i})^2}{1+\mathrm{i}} = \frac{-2\mathrm{i}}{1+\mathrm{i}} \times \frac{1-\mathrm{i}}{1-\mathrm{i}} = \frac{-2\mathrm{i}-2}{1+1} = -1-\mathrm{i}$$

36) g)

$$\frac{a+b\mathrm{i}}{b-a\mathrm{i}} \times \frac{b+a\mathrm{i}}{b+a\mathrm{i}} = \frac{ab+a^2\mathrm{i}+b^2\mathrm{i}-ab}{b^2+a^2} = \frac{a^2\mathrm{i}+b^2\mathrm{i}}{b^2+a^2} = \frac{(a^2+b^2)\mathrm{i}}{b^2+a^2} = \mathrm{i}$$

37)

$$\frac{5}{z} = \frac{5}{-1+2\mathrm{i}} \times \frac{-1-2\mathrm{i}}{-1-2\mathrm{i}} = \frac{-5-10\mathrm{i}}{1+4} = -1-2\mathrm{i}$$

$$z + \frac{5}{z} = -1+2\mathrm{i} + \frac{5}{-1+2\mathrm{i}} = -1+2\mathrm{i}-1-2\mathrm{i} = -2$$

38) a)

$$(3-2\mathrm{i})(4+\mathrm{i}) = 12-5\mathrm{i}+2 = (14-5\mathrm{i}) \text{ volts}$$

38) b)

$$\frac{6-2\mathrm{i}}{2+3\mathrm{i}} \times \frac{2-3\mathrm{i}}{2-3\mathrm{i}} = \frac{12-18\mathrm{i}-4\mathrm{i}-6}{4+9} = \left(\frac{6}{13} - \frac{22}{13}\mathrm{i}\right) \text{ ohms}$$

39)

$$\frac{15-5\mathrm{i}}{z} = 2-3\mathrm{i} \Rightarrow z = \frac{15-5\mathrm{i}}{2-3\mathrm{i}} \times \frac{2+3\mathrm{i}}{2+3\mathrm{i}} = \frac{30-10\mathrm{i}+45\mathrm{i}+15}{4+9} = \frac{45}{13} + \frac{35}{13}\mathrm{i}$$

40)

$\frac{2}{3+\mathrm{i}} \times \frac{3-\mathrm{i}}{3-\mathrm{i}} = \frac{6-2\mathrm{i}}{9+1} = \frac{3}{5} - \frac{1}{5}\mathrm{i}$ and $\frac{3}{2+\mathrm{i}} \times \frac{2-\mathrm{i}}{2-\mathrm{i}} = \frac{6-3\mathrm{i}}{4+1} = \frac{6}{5} - \frac{3}{5}\mathrm{i}$

So $\frac{2}{3+\mathrm{i}} + \frac{3}{2+\mathrm{i}} = \frac{3}{5} - \frac{1}{5}\mathrm{i} + \frac{6}{5} - \frac{3}{5}\mathrm{i} = \frac{9}{5} - \frac{4}{5}\mathrm{i}$

41)

$\frac{1}{\cos\theta - \mathrm{i}\sin\theta} \times \frac{\cos\theta + \mathrm{i}\sin\theta}{\cos\theta + \mathrm{i}\sin\theta} = \frac{\cos\theta + \mathrm{i}\sin\theta}{\cos^2\theta + \sin^2\theta} = \cos\theta + \mathrm{i}\sin\theta$

42) a)

$\frac{1}{x+\mathrm{i}y} \times \frac{x-\mathrm{i}y}{x-\mathrm{i}y} = \frac{x-\mathrm{i}y}{x^2+y^2} = \frac{x}{x^2+y^2} - \frac{y}{x^2+y^2}\mathrm{i}$

$\therefore \mathrm{Re}\left(x + \mathrm{i}y + \frac{1}{x+\mathrm{i}y}\right) = x + \frac{x}{x^2+y^2}$ and $\mathrm{Im}\left(x + \mathrm{i}y + \frac{1}{x+\mathrm{i}y}\right) = y - \frac{y}{x^2+y^2}$

42) b)

$-\frac{1}{x+\mathrm{i}y} \times \frac{x-\mathrm{i}y}{x-\mathrm{i}y} = -\frac{x-\mathrm{i}y}{x^2+y^2} = -\frac{x}{x^2+y^2} + \frac{y}{x^2+y^2}\mathrm{i}$

$\therefore \mathrm{Re}\left(x + \mathrm{i}y - \frac{1}{x+\mathrm{i}y}\right) = x - \frac{x}{x^2+y^2}$ and $\mathrm{Im}\left(x + \mathrm{i}y - \frac{1}{x+\mathrm{i}y}\right) = y + \frac{y}{x^2+y^2}$

42) c)

$\frac{1}{x-\mathrm{i}y} \times \frac{x+\mathrm{i}y}{x+\mathrm{i}y} = \frac{x+\mathrm{i}y}{x^2+y^2} = \frac{x}{x^2+y^2} + \frac{y}{x^2+y^2}\mathrm{i}$

$\therefore \mathrm{Re}\left(x - \mathrm{i}y + \frac{1}{x-\mathrm{i}y}\right) = x + \frac{x}{x^2+y^2}$ and $\mathrm{Im}\left(x - \mathrm{i}y + \frac{1}{x-\mathrm{i}y}\right) = -y + \frac{y}{x^2+y^2}$

42) d)

$-\frac{1}{x-\mathrm{i}y} \times \frac{x+\mathrm{i}y}{x+\mathrm{i}y} = -\frac{x+\mathrm{i}y}{x^2+y^2} = -\frac{x}{x^2+y^2} - \frac{y}{x^2+y^2}\mathrm{i}$

$\therefore \mathrm{Re}\left(x - \mathrm{i}y - \frac{1}{x-\mathrm{i}y}\right) = x - \frac{x}{x^2+y^2}$ and $\mathrm{Im}\left(x - \mathrm{i}y - \frac{1}{x-\mathrm{i}y}\right) = -y - \frac{y}{x^2+y^2}$

43)

$\frac{1+\mathrm{i}}{1-\mathrm{i}} = \frac{1+\mathrm{i}}{1-\mathrm{i}} \times \frac{1+\mathrm{i}}{1+\mathrm{i}} = \frac{1+2\mathrm{i}-1}{1+1} = \mathrm{i}$

So least positive integer n such that $\mathrm{i}^n = 1$

$\mathrm{i}^2 = -1$, $\mathrm{i} = -\mathrm{i}$, $\mathrm{i}^4 = 1 \therefore n = 4$

44)

$\frac{a(1-2\mathrm{i})(3-2\mathrm{i})}{1-2\mathrm{i}}+\frac{b(1-2\mathrm{i})(3-2\mathrm{i})}{3-2\mathrm{i}}=\frac{-5+6\mathrm{i}}{1+8\mathrm{i}}\times(1-2\mathrm{i})(3-2\mathrm{i})$

$a(3-2\mathrm{i})+b(1-2\mathrm{i})=\frac{-5+6\mathrm{i}}{1+8\mathrm{i}}\times(-1-8\mathrm{i})$

$(3a+b)+(-2a-2b)\mathrm{i}=5-6\mathrm{i}$

$3a+b=5$ and $-2a-2b=-6$

Which gives $a=1$, $b=2$

Alternative Method:

$\frac{a}{1-2\mathrm{i}}\times\frac{1+2\mathrm{i}}{1+2\mathrm{i}}=\frac{a+2a\mathrm{i}}{1+4}=\frac{a}{5}+\frac{2a}{5}\mathrm{i}$

$\frac{b}{3-2\mathrm{i}}\times\frac{3+2\mathrm{i}}{3+2\mathrm{i}}=\frac{3b+2b\mathrm{i}}{9+4}=\frac{3b}{13}+\frac{2b}{13}\mathrm{i}$

$\frac{-5+6\mathrm{i}}{1+8\mathrm{i}}\times\frac{1-8\mathrm{i}}{1-8\mathrm{i}}=\frac{-5+40\mathrm{i}+6\mathrm{i}+48}{1+64}=\frac{43}{65}+\frac{46}{65}\mathrm{i}$

$\frac{a}{5}+\frac{3b}{13}=\frac{43}{65}$ and $\frac{2a}{5}+\frac{2b}{13}=\frac{46}{65}$

Solving simultaneously: $a=1$, $b=2$

45)

$zz^*=(a+b\mathrm{i})(a-b\mathrm{i})=a^2+b^2=17$

$\frac{z}{z^*}=\frac{a+b\mathrm{i}}{a-b\mathrm{i}}\times\frac{a+b\mathrm{i}}{a+b\mathrm{i}}=\frac{a^2-b^2+2ab\mathrm{i}}{a^2+b^2}=\frac{a^2-b^2}{a^2+b^2}+\frac{2ab}{a^2+b^2}\mathrm{i}$ so that $\frac{a^2-b^2}{a^2+b^2}=\frac{15}{17}$ and $\frac{2ab}{a^2+b^2}=-\frac{8}{17}$

So $a^2-b^2=15$ and $2ab=-8$

Solve simultaneously:

$b=-\frac{4}{a}$ gives $a^2-\left(-\frac{4}{a}\right)^2=15\Rightarrow a^4-16=15a^2\Rightarrow a^4-15a^2-16=0$

$a^2=\frac{15\pm\sqrt{225+64}}{2}=\frac{15}{2}\pm\frac{17}{2}$ so $a^2=16$ or $a^2=-1$

Since a is real, $a=4$ or $a=-4$

$\therefore a=4, b=-1$ or $a=-4, b=1$

46)

Let $z=a+b\mathrm{i}$

$\frac{z}{z-2}=\frac{a+b\mathrm{i}}{a-2+b\mathrm{i}}\times\frac{a-2-b\mathrm{i}}{a-2-b\mathrm{i}}=\frac{a^2-2a-ab\mathrm{i}+ab\mathrm{i}-2b\mathrm{i}+b^2}{(a-2)^2+b^2}=\frac{a^2-2a-2b\mathrm{i}+b^2}{(a-2)^2+b^2}=\frac{a^2-2a+b^2}{(a-2)^2+b^2}+\frac{-2b}{(a-2)^2+b^2}\mathrm{i}$

$\mathrm{Re}\left(\frac{z}{z-2}\right)=\frac{a^2-2a+b^2}{(a-2)^2+b^2}$

$\mathrm{Im}\left(\frac{z}{z-2}\right)=\frac{-2b}{(a-2)^2+b^2}$

47)

$\frac{z}{3+4\mathrm{i}} = \frac{a+b\mathrm{i}}{3+4\mathrm{i}} \times \frac{3-4\mathrm{i}}{3-4\mathrm{i}} = \frac{3a-4a\mathrm{i}+3b\mathrm{i}+4b}{9+16} = \frac{3a+4b}{25} + \frac{-4a+3b}{25}\mathrm{i}$

$\frac{z-1}{5\mathrm{i}} = \frac{a+b\mathrm{i}-1}{5\mathrm{i}} \times \frac{-5\mathrm{i}}{-5\mathrm{i}} = \frac{-5a\mathrm{i}+5b+5\mathrm{i}}{25} = \frac{5b}{25} + \frac{-5a+5}{25}\mathrm{i}$

$\frac{5}{3-4\mathrm{i}} \times \frac{3+4\mathrm{i}}{3+4\mathrm{i}} = \frac{15+20\mathrm{i}}{9+16} = \frac{15}{25} + \frac{20}{25}\mathrm{i}$

Equating real and imaginary parts:

$\frac{3a+4b}{25} + \frac{5b}{25} = \frac{15}{25}$ and $\frac{-4a+3b}{25} + \frac{-5a+5}{25} = \frac{20}{25}$

$3a + 9b = 15$ and $-9a + 3b + 5 = 20$

$\therefore a = -1, b = 2$

$\therefore z = -1 + 2\mathrm{i}$

48)

$\frac{a-b\mathrm{i}}{c-d\mathrm{i}} = \frac{a-b\mathrm{i}}{c-d\mathrm{i}} \times \frac{c+d\mathrm{i}}{c+d\mathrm{i}} = \frac{ac-bc\mathrm{i}+ad\mathrm{i}+bd}{c^2+d^2} = \frac{ac+bd}{c^2+d^2} - \frac{bc-ad}{c^2+d^2}\mathrm{i}$

$(x - \mathrm{i}y)^2 = \frac{a-b\mathrm{i}}{c+d\mathrm{i}} = \frac{ac+bd}{c^2+d^2} - \frac{bc-ad}{c^2+d^2}\mathrm{i}$

$(x - \mathrm{i}y)^2 = (x - \mathrm{i}y)(x - \mathrm{i}y) = x^2 - y^2 - 2xy\mathrm{i}$

Compare real components:

$x^2 - y^2 = \frac{ac+bd}{c^2+d^2}$ as req.